

VPC & Networking

Subnets · Security Groups · NAT · Endpoints · Transit Gateway

AWS Tutorial · ShopFlow

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Chapter Overview

Topic	AWS Service	ShopFlow Use-Case
VPC Fundamentals	VPC, CIDR blocks, subnets	Isolate prod network 10.0.0.0/16
Subnet Design	Public, Private App, Private Data	3-tier security isolation
Security Groups	Stateful instance firewall	Allow ALB→ECS, ECS→RDS only
NACLs	Stateless subnet firewall	Block entire IP ranges at subnet level
Internet & NAT Gateway	IGW, NAT Gateway	Outbound-only internet for ECS tasks
VPC Peering & Transit Gateway	VPC Peering, TGW	Connect prod and shared-services VPCs
VPC Endpoints	Gateway, Interface endpoints	Private S3/DynamoDB access, no internet
Route Tables	Routing	Control traffic flow between subnets

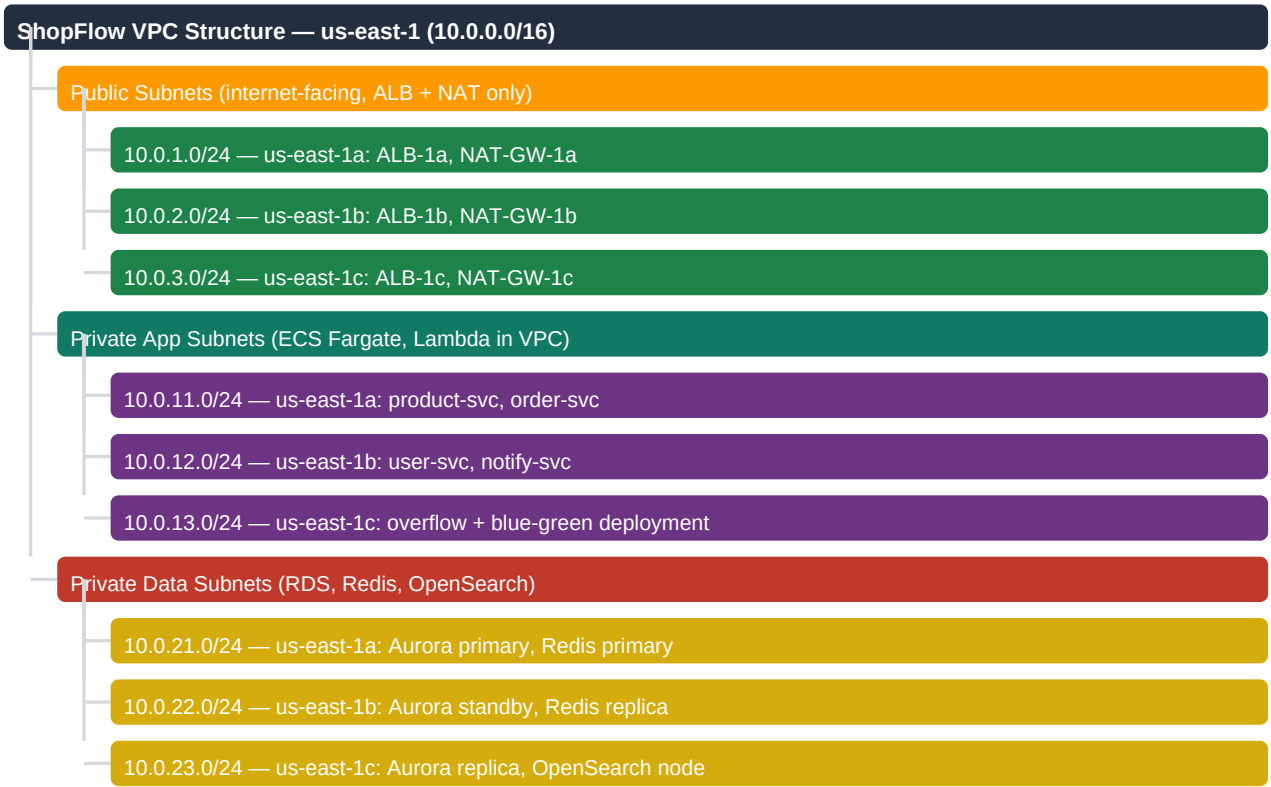
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VPC Fundamentals — Your Private Cloud Network

A Virtual Private Cloud (VPC) is a logically isolated section of the AWS cloud where you launch resources in a virtual network you define. Every VPC is region-scoped — it spans all AZs in a Region. Without a VPC, your resources would be on a flat public network accessible to everyone.

VPC Key Concepts

Concept	Description	ShopFlow Value
CIDR Block	IP range assigned to the VPC (e.g., 10.0.0.0/16)	10.0.0.0/16 = 65,536 IPs gives 65K IPs for growth
Subnets	Subdivisions of VPC CIDR, each in one AZ	27 subnets across 3 AZs (3 tiers x 3 AZs)
Route Tables	Controls where traffic is directed (local, IGW, NAT, VPC)	Public: default route to IGW; Private: to NAT
Internet Gateway (IGW)	Enables internet access for resources with public IPs attached to VPC, used by ALB in public subnets	App
NAT Gateway	Allows private subnet outbound internet (no inbound)	ECS tasks download Docker images, call Stripe
Security Groups	Stateful firewall at instance/ENI level	Layer 3 of ShopFlow's network isolation
NACLs	Stateless firewall at subnet level	Block known-bad IP ranges at perimeter



■ Size your VPC CIDR for 10x your expected workload. ShopFlow uses /16 (65,536 IPs). Each /24 subnet gives 251 usable IPs (AWS reserves 5 per subnet). Never use /28 or smaller for production.

03-02 Subnet Design — 3-Tier Architecture

ShopFlow uses a 3-tier subnet design that provides defense-in-depth. Each tier has a different trust level and different routing. This architecture makes lateral movement by an attacker much harder.

3-Tier Security Model

Tier	Subnet Name	Resources Placed Here	Internet Access	Access Restrictions
Tier 1 — Public	public-1a/1b/1c	ALB, NAT Gateways, Bastion (temp)	Inbound + Outbound via IGW	Security Group limits to 80/443 only
Tier 2 — App (Private)	app-1a/1b/1c	ECS Fargate tasks, Lambda in VPC	Outbound only via NAT GW (only placed)	Only accessible from ALB Security Groups
Tier 3 — Data (Private)	data-1a/1b/1c	RDS Aurora, ElastiCache, OpenSearch	None — no route to internet at all	Only accessible from App subnet SGs

Route Table Configuration

Route Table	Subnet	Destination	Target
public-rt	public-1a/1b/1c	0.0.0.0/0 (internet)	Internet Gateway (igw-xxx)
public-rt	public-1a/1b/1c	10.0.0.0/16 (VPC local)	local (automatic)
app-rt-1a	app-1a	0.0.0.0/0 (internet)	NAT Gateway in public-1a
app-rt-1a	app-1a	10.0.0.0/16 (VPC local)	local (automatic)
app-rt-1b	app-1b	0.0.0.0/0 (internet)	NAT Gateway in public-1b
data-rt	data-1a/1b/1c	10.0.0.0/16 (VPC local)	local only — no internet route!

CDK VPC Setup

```
const vpc = new ec2.Vpc(this, "ShopFlowVpc", {
  ipAddresses: ec2.IpAddresses.cidr("10.0.0.0/16"),
  maxAzs: 3,
  natGateways: 3, // one per AZ for HA
  subnetConfiguration: [
    {name:"public", subnetType:ec2.SubnetType.PUBLIC, cidrMask:24},
    {name:"app", subnetType:ec2.SubnetType.PRIVATE_WITH_EGRESS, cidrMask:24},
    {name:"data", subnetType:ec2.SubnetType.PRIVATE_ISOLATED, cidrMask:24},
  ],
});
```

■ NAT Gateway costs \$0.045/hr per gateway + \$0.045/GB processed. For 3 AZs = \$97/month just for NAT. Use VPC Endpoints for S3/DynamoDB access to avoid data charges through NAT.

Security Groups are the primary network firewall for EC2, ECS tasks, Lambda in VPC, RDS, ElastiCache, and ALBs. They are STATEFUL — if an inbound connection is allowed, the return traffic is automatically allowed without an explicit outbound rule.

ShopFlow Security Group Design

Security Group	Inbound Rules	Outbound Rules	Attached To
sg-alb	443/tcp from 0.0.0.0/0 (internet) 80/tcp from 0.0.0.0/0 (redirect)	8080/tcp to sg-ecs-app	Application Load Balancer
sg-ecs-app	8080/tcp from sg-alb only	5432/tcp to sg-rds 6379/tcp to sg-redis 443/tcp to 0.0.0.0/0 (Stripe, etc.)	ECS Fargate tasks
sg-rds	5432/tcp from sg-ecs-app only	None needed (stateful)	RDS Aurora PostgreSQL
sg-redis	6379/tcp from sg-ecs-app only	None needed (stateful)	ElastiCache Redis
sg-opensearch	443/tcp from sg-ecs-app only	None needed (stateful)	OpenSearch domain

Security Group — CDK Implementation

```
// ALB Security Group
const sgAlb = new ec2.SecurityGroup(this, "SgAlb", {
  vpc, description: "ALB — internet-facing HTTPS",
  allowAllOutbound: false,
});
sgAlb.addIngressRule(ec2.Peer.anyIpv4(), ec2.Port.tcp(443), "HTTPS");
sgAlb.addIngressRule(ec2.Peer.anyIpv4(), ec2.Port.tcp(80), "HTTP redirect");
// ECS App Security Group
const sgApp = new ec2.SecurityGroup(this, "SgApp", {
  vpc, description: "ECS tasks — private app tier",
  allowAllOutbound: false,
});
sgApp.addIngressRule(sgAlb, ec2.Port.tcp(8080), "From ALB");
sgApp.addEgressRule(sgRds, ec2.Port.tcp(5432), "To Aurora");
sgApp.addEgressRule(sgRedis, ec2.Port.tcp(6379), "To Redis");
sgApp.addEgressRule(ec2.Peer.anyIpv4(), ec2.Port.tcp(443), "Ext APIs");
```

■ Reference Security Groups in rules (not CIDR blocks) for dynamic scaling. When a new ECS task starts in sg-ecs-app, it automatically inherits all the correct firewall rules.

Network Access Control Lists (NACLs) are an additional, stateless firewall layer at the subnet boundary. Because they are stateless, you must explicitly allow BOTH inbound AND outbound traffic for each connection (including ephemeral return ports 1024-65535).

NACLs vs Security Groups

Feature	NACLs	Security Groups
Stateful/Stateless	Stateless — must allow both directions	Stateful — return traffic auto-allowed
Level	Subnet level	Instance/ENI level
Rules	Numbered rules, evaluated in order (lowest first)	All rules evaluated simultaneously
Default action	Rule 32767 DENY ALL (implicit)	Implicit deny (no explicit rule needed)
Allow & Deny	Can explicitly DENY and ALLOW	Allow only — no deny rules

Feature	NACLs	Security Groups
Use case	Block IP ranges, extra subnet perimeter	Primary firewall for resources
ShopFlow usage	Block known-bad IPs, emergency IP blocks	Primary firewall — all rules here

ShopFlow Public Subnet NACL

Rule	Type	Protocol	Port Range	Source	Action
100	Inbound	TCP	443	0.0.0.0/0	ALLOW (HTTPS)
110	Inbound	TCP	80	0.0.0.0/0	ALLOW (HTTP redirect)
120	Inbound	TCP	1024-65535	0.0.0.0/0	ALLOW (return traffic)
200	Inbound	TCP	any	192.0.2.0/24	DENY (known-bad IP)
* ALL ALL ALL ALL DENY (implicit)					
100	Outbound	TCP	8080	10.0.11.0/22	ALLOW (to ECS)
110	Outbound	TCP	1024-65535	0.0.0.0/0	ALLOW (return to clients)
* ALL ALL ALL ALL DENY (implicit)					

■ Use NACLs sparingly — they add complexity. Use Security Groups for primary access control. Use NACLs only for emergency IP blocking or compliance requirements that need explicit subnet-level deny.

The Internet Gateway (IGW) enables two-way internet traffic for public subnets. NAT Gateways enable outbound-only internet traffic from private subnets. In ShopFlow, only the ALB is internet-facing — ECS tasks are private and use NAT for outbound calls to Stripe, SendGrid, etc.

IGW vs NAT Gateway

Feature	Internet Gateway (IGW)	NAT Gateway
Traffic direction	Inbound + Outbound	Outbound only (no inbound)
Requires public IP	Yes — instance needs Elastic IP or auto-assigned public IP	No — public resources keep private IPs
Managed by	AWS (fully managed, no patching)	AWS (fully managed, auto-scaled)
Cost	Free	\$0.045/hr + \$0.045/GB processed per AZ
High Availability	One per VPC, redundant by design	One per AZ (create 3 for HA)
Use case	ALB, public-facing resources	ECS tasks pulling images, calling ext APIs

NAT Gateway HA Design — ShopFlow

ShopFlow deploys one NAT Gateway per AZ. If AZ 1a fails and ECS tasks in 1a route to NAT in 1b, cross-AZ data charges apply and NAT becomes a bottleneck. One NAT per AZ is the correct production pattern.

```
# CDK: natGateways: 3 creates one NAT GW per AZ automatically
# Each private subnet in an AZ routes to the NAT GW in that AZ
# Verify route tables after CDK deploy:
aws ec2 describe-route-tables \
  --filters Name=tag:Name,Values=ShopFlowVpc/appSubnet1a \
  --query "RouteTables[].Routes"
# Expected: 0.0.0.0/0 → nat-0123abc (NAT GW in 1a)
```

NAT Gateway Cost Optimization

- VPC Endpoints (next section) eliminate NAT GW data charges for S3/DynamoDB/SQS
- Cost breakdown: \$0.045/hr = \$32/month per NAT GW. For 3 AZs = \$97/month base
- Plus \$0.045/GB processed: 1TB/month = \$46. Total without endpoints: ~\$143/month
- With VPC endpoints for S3 (most traffic): saves ~\$40/month in data charges

■ Monitor NAT Gateway CloudWatch metric BytesOutToDestination to understand traffic volume. If throughput is consistently high, check if VPC Endpoints can handle those destinations instead.

VPC Endpoints allow private connectivity between your VPC and AWS services without routing traffic through the internet or NAT Gateway. This improves security (traffic never leaves AWS backbone), performance, and reduces costs.

VPC Endpoint Types

Endpoint Type	Services	Cost	How It Works	ShopFlow Usage
Gateway Endpoint	S3, DynamoDB only	Free	Adds prefix list to route table	S3 (images), DynamoDB (cart)
Interface Endpoint (EC2 private IP)	Most services: SQS, SNS, Secrets Manager, ECR, CloudWatch, etc.	\$0.0146/AZ-CPU-01/GWates, etc.	Creates ENI in your subnet with private IP	SQS, Secrets Manager, ECR, Cloud
Gateway Load Balancer Endpoint	Third-party appliances (firewalls, IDS)	\$1.00/hour	Routes traffic through appliance fleet	Not used in ShopFlow

ShopFlow VPC Endpoints Setup

```
# CDK: Add Gateway Endpoints (free)
```

```
vpc.addGatewayEndpoint("S3Endpoint", {
  service: ec2.GatewayVpcEndpointAwsService.S3,
});
vpc.addGatewayEndpoint("DynamoDbEndpoint", {
  service: ec2.GatewayVpcEndpointAwsService.DYNAMODB,
});
# CDK: Add Interface Endpoints (paid)
vpc.addInterfaceEndpoint("SqsEndpoint", {
  service: ec2.InterfaceVpcEndpointAwsService.SQS,
  subnets: {subnetType: ec2.SubnetType.PRIVATE_WITH_EGRESS},
  privateDnsEnabled: true,
});
vpc.addInterfaceEndpoint("SecretsManagerEndpoint", {
  service: ec2.InterfaceVpcEndpointAwsService.SECRETS_MANAGER,
  subnets: {subnetType: ec2.SubnetType.PRIVATE_WITH_EGRESS},
});
```

VPC Endpoint	ShopFlow Traffic Routed	Cost Saving vs NAT
S3 Gateway Endpoint	Product images reads (ECS→S3), exports	\$40-80/month (free endpoint, avoids NAT data charge)
DynamoDB Gateway Endpoint	Cart reads/writes (ECS→DynamoDB)	\$15-30/month
SQS Interface Endpoint	Order event publishing (ECS→SQS)	\$8/month (vs NAT) — break even at ~200GB/month
Secrets Manager Interface Endpoint	Secret retrieval on task startup	Small but adds security (traffic off internet)
ECR Interface Endpoint	Docker image pulls at task launch	\$20-40/month (large images pulled frequently)

■ Gateway Endpoints for S3 and DynamoDB are always free — add them to every VPC. Interface Endpoints cost \$7.20/month per AZ; cost-justify based on NAT data charges saved.

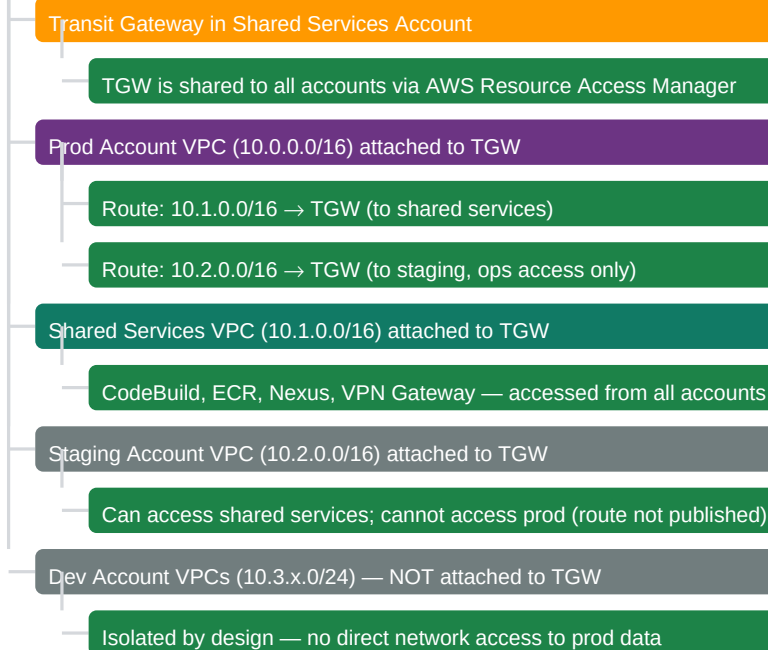
ShopFlow has multiple VPCs (prod, staging, shared-services) that need to communicate privately. VPC Peering works for simple point-to-point connections. Transit Gateway handles complex hub-and-spoke topologies.

VPC Peering vs Transit Gateway

Feature	VPC Peering	Transit Gateway (TGW)
Topology	One-to-one only — no transitive routing	Hub-and-spoke — transitive routing
Max connections	125 peering connections per VPC	5,000 VPC attachments per TGW
Bandwidth	No limit, full VPC bandwidth	50 Gbps per VPC attachment
Cost	\$0.01/GB data transfer (cross-AZ)	\$0.05/hr + \$0.02/GB processed
Cross-region	Yes, inter-region peering supported	Yes, via TGW peering
Cross-account	Yes, requester/acceptor model	Yes, using RAM (Resource Access Manager)
Use case	2-3 VPCs, simple topology	5+ VPCs, complex routing, centralized egress

ShopFlow Network Topology

ShopFlow Multi-Account Network (Transit Gateway)



■ VPC Peering is NOT transitive. If A↔B and B↔C, A cannot reach C. Use Transit Gateway when you have 3+ VPCs that all need to communicate.

VPC Flow Logs capture metadata about network traffic to and from your VPC resources. They are essential for security incident investigation, compliance, and network optimization. They log IP addresses, ports, protocols, packet counts, and accept/reject decisions.

Flow Log Configuration — ShopFlow

```
# CDK: Enable VPC Flow Logs to CloudWatch and S3
```



```
new ec2.FlowLog(this, "VpcFlowLogCW", {
  resourceType: ec2.FlowLogResourceType.fromVpc(vpc),
  destination: ec2.FlowLogDestination.toCloudWatchLogs(
    new logs.LogGroup(this, "FlowLogGroup", {
      retention: logs.RetentionDays.THIRTY_DAYS,
    })
  ),
  trafficType: ec2.FlowLogTrafficType.ALL,
});

# Flow log format: version, account, interface, src/dst IPs,
# ports, protocol, packets, bytes, start/end, action, status
2 123456789012 eni-abc123 10.0.11.5 10.0.21.10 52341 5432 6 10 840 \
1609459200 1609459260 ACCEPT OK
```

Security Analysis with Flow Logs

Investigation	CloudWatch Insights Query	Alert On
Port scan detection	filter action="REJECT" stats count(*) by srcAddr	Any IP rejected 1000+ times in 5 min
Data exfiltration	filter dstPort=443 stats sum(bytes) by srcAddr	Any single IP sending >1GB/hr
Unauthorized DB access	filter dstPort=5432 AND srcAddr NOT IN 10.0.10.0/24	Any DB connection from non-app subnet
Unexpected outbound	filter srcAddr=10.0.21.0/24 AND dstAddr NOT IN 10.0.10.0/24	Any outbound from data subnet

■ Enable flow logs on ALL traffic (ACCEPT + REJECT), not just REJECT. Accepted traffic analysis reveals unexpected communication paths and is needed for lateral movement investigation.

If ShopFlow needs to connect its on-premises data center (for legacy systems, analytics tools, or hybrid architecture), AWS offers AWS Direct Connect for dedicated private connectivity and Site-to-Site VPN for encrypted internet-based connectivity.

Feature	AWS Direct Connect	Site-to-Site VPN
Bandwidth	1/10/100 Gbps dedicated fiber	Up to 1.25 Gbps per tunnel (2 tunnels/VPN)
Latency	Consistent, low — dedicated link	Variable, depends on internet path
Encryption	Not encrypted by default (private link)	IPSec encrypted by default
Setup time	Weeks (physical fiber provisioning)	Minutes (software config)
Cost	\$0.025-0.30/GB + \$0.30/hr port	\$0.05/hr + \$0.01/GB
Reliability	SLA 99.99% (with redundant connections)	Internet reliability (99.9%)
ShopFlow use	Analytics from on-prem Hadoop cluster	Dev team VPN access to prod VPC

Site-to-Site VPN — Quick Setup

1. Create Customer Gateway (represents your on-prem router)

```
aws ec2 create-customer-gateway \
--type ipsec.1 \
--public-ip 203.0.113.1 \ # on-prem router public IP
--bgp-asn 65000
```

2. Create Virtual Private Gateway and attach to VPC

```
aws ec2 create-vpn-gateway --type ipsec.1
aws ec2 attach-vpn-gateway --vpn-gateway-id vgw-xxx --vpc-id vpc-xxx
```

3. Create VPN Connection

```
aws ec2 create-vpn-connection \
--type ipsec.1 \
--customer-gateway-id cgw-xxx \
--vpn-gateway-id vgw-xxx
```

Download tunnel config for your router (Cisco, Juniper, pfSense, etc.)

■ Always configure two VPN tunnels for HA. Each tunnel terminates in a different AWS endpoint. If one tunnel fails, BGP routing automatically switches to the second tunnel in <30 seconds.

In ShopFlow's production VPC, there are no SSH keys and no port 22 open anywhere. Instead of a traditional bastion host, ShopFlow uses AWS Systems Manager Session Manager — no SSH, no open ports, full audit log, no key management.

Session Manager vs Bastion Host

Feature	Traditional Bastion Host	AWS SSM Session Manager
Port requirement	Port 22 open (attack surface)	Zero open ports (uses HTTPS internally)
Authentication	SSH key (can be stolen/lost)	IAM permissions + MFA (controlled centrally)
Audit log	SSH logs (often incomplete)	Full session recording to S3 + CloudTrail
Maintenance	OS patching, HA setup, costs \$50-100/month	Fully managed by AWS, no servers to run
Emergency access	Need SSH key in hand	Break-glass IAM role, usable from browser
ShopFlow usage	Eliminated completely	All debug sessions via Session Manager

Session Manager Setup — ShopFlow

```
# 1. Attach SSM policy to ECS task (or EC2 instance role)
taskRole.addManagedPolicy(
ManagedPolicy.fromAwsManagedPolicyName(
"AmazonSSMManagedInstanceCore"));

# 2. Start interactive session from CLI (no SSH key needed)
aws ssm start-session \
--target ecs:shopflow_prod_product-svc-container \
--profile shopflow-prod-readonly

# 3. Run command non-interactively
aws ssm send-command \
--targets Key=tag:Service,Values=product-svc \
--document-name "AWS-RunShellScript" \
--parameters commands=["curl localhost:8080/actuator/health"]
```

■ ECS Exec (built on SSM) allows exec into running containers without SSH: `aws ecs execute-command --cluster shopflow --task TASK_ID --command /bin/sh --interactive`. Zero SSH keys, full CloudTrail audit.

AWS now supports IPv6 natively in VPCs. ShopFlow's API layer is dual-stack (IPv4 + IPv6) to support mobile carriers that prefer IPv6 and to future-proof the architecture.

Feature	IPv4	IPv6 in AWS
Address space	~4.3B addresses (exhausted publicly)	340 undecillion — no shortage
Private addresses	10.x.x.x / 172.16.x.x / 192.168.x.x	No private range — all addresses are unique
AWS VPC CIDR	You choose (e.g., 10.0.0.0/16)	AWS assigns a /56 per VPC automatically
NAT required for private	Yes — NAT Gateway for internet	No — Egress-Only IGW for outbound only
Cost	Standard	IPv6 addresses: free; data transfer: standard
ShopFlow adoption	All internal traffic	ALB + CloudFront dual-stack for mobile users

■ Enable IPv6 on ALB and CloudFront distributions. Mobile carriers (especially in Asia and Africa) increasingly default to IPv6. Dual-stack eliminates connection failures for pure-IPv6 clients.

ShopFlow Production Network Architecture

Internet → CloudFront (edge, 600+ PoPs globally)

WAF: OWASP rules, rate limiting (2000 req/5min per IP)

TLS 1.2+ via ACM certificate (auto-renewed)

us-east-1 Region → VPC 10.0.0.0/16

IGW attached to VPC (enables public subnet internet access)

Public Subnets (10.0.1-3.0/24) — 3 AZs

ALB: listeners 443 (HTTPS) + 80→443 redirect

NAT Gateways: 3x (one per AZ) for private subnet outbound

Private App Subnets (10.0.11-13.0/24) — 3 AZs

ECS Fargate: 5 services, Security Group allows 8080 from ALB only

Lambda@VPC: notification processors, secret-rotation helpers

VPC Endpoint: SQS, Secrets Manager, ECR (interface endpoints)

Private Data Subnets (10.0.21-23.0/24) — NO internet route

Aurora PostgreSQL: SG allows 5432 from sg-ecs-app only

ElastiCache Redis: SG allows 6379 from sg-ecs-app only

OpenSearch: SG allows 443 from sg-ecs-app only

VPC Endpoints: S3 (gateway, free), DynamoDB (gateway, free)

Transit Gateway: peered to Shared Services VPC (10.1.0.0/16)

Route: 10.1.0.0/16 → TGW (ECR, CodePipeline, Nexus)

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Best Practices & Anti-Patterns

- ✓ Use 3-tier subnet design (Public/App/Data) from Day 1 — retrofitting is painful
- ✓ Add Security Group references (not CIDR) in SG rules for dynamic scaling
- ✓ Deploy one NAT Gateway per AZ to avoid cross-AZ traffic during AZ failure
- ✓ Add Gateway VPC Endpoints for S3 and DynamoDB — they are free and save money
- ✓ Enable VPC Flow Logs on ALL traffic (accept + reject) for security investigation
- ✓ Use SSM Session Manager instead of SSH/bastion for all debug access
- ✓ Size VPC CIDR generously (/16) — you cannot resize a VPC after creation
- Never allow 0.0.0.0/0 inbound on port 22, 3389, 5432, 3306, 6379, 9200
- Never place RDS, ElastiCache, or OpenSearch in public subnets
- Do not use a single NAT Gateway for all AZs — single point of failure
- Do not use overlapping CIDR ranges if you plan VPC peering or Direct Connect

■ NACLs are stateless — always allow ephemeral ports (1024-65535) for return traffic

Concept	Key Fact
VPC scope	Region-wide, spans all AZs in a Region
Subnet scope	Single AZ only — each subnet is in one AZ
IGW purpose	Two-way internet for public subnets (free)
NAT Gateway purpose	Outbound-only internet for private subnets
NAT Gateway cost	\$0.045/hr + \$0.045/GB processed per AZ
Security Group state	Stateful — return traffic auto-allowed
NACL state	Stateless — must explicitly allow both directions
VPC Endpoint Gateway	S3 and DynamoDB only — FREE
VPC Endpoint Interface	100+ services — \$0.01/hr/AZ + \$0.01/GB
VPC Peering transitive?	No — A↔B and B↔C does NOT mean A↔C
AWS reserved IPs/subnet	5 IPs per subnet (first 4 + broadcast)
Flow Logs destination	CloudWatch Logs or S3 (or both)

Top Interview Questions

Question	Answer
What is a VPC and why is it needed?	Logically isolated virtual network in AWS. Without VPC, resources are on a flat public network.
Difference between Security Groups and NACLs?	SG: stateful, instance-level, allow only, all rules evaluated. NACL: stateless, subnet-level, allow/deny both.
What is a NAT Gateway and when needed?	Allows resources in private subnets to initiate outbound internet connections (e.g., pull Docker images).
VPC Peering vs Transit Gateway?	Peering: direct 1-to-1, no transitive routing, max 125 connections, cheapest. TGW: hub-and-spoke, multi-region, more expensive.
How do you access S3 from a private subnet?	Amazon S3 Gateway VPC Endpoint (free). This adds S3 as a route table entry — traffic to S3 goes through endpoint.
What are VPC Flow Logs?	Capture metadata on network flows: src/dst IP, ports, protocol, bytes, action (ACCEPT/REJECT).
How do you debug ECS containers with AWS CLI?	aws ecs exec (built on SSM Session Manager): aws ecs execute-command. No open ports needed.